

Demand over capacity

Congestion is a ratio: demand divided by capacity

The idea

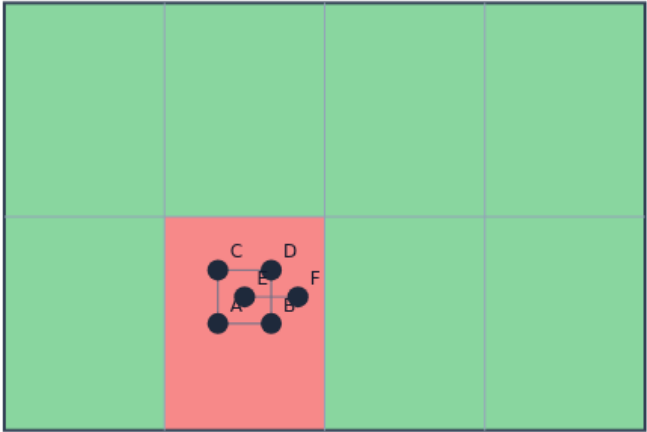
- $\text{Cong}[i][j]$ equals $\text{demand}[i][j]$ over Cap
- Values above one are oversubscribed
- The hottest GCell is the argmax
- On the congested seed, expect the center columns to light up first

Demand / Cap

CONGESTION MAP

Demand / Cap

Step 1 / 5 · ratio · module02-05-congestion-map



Explanation

Congestion is a ratio per GCell—values above one are oversubscribed.

- Heat = ratio

Cap=2

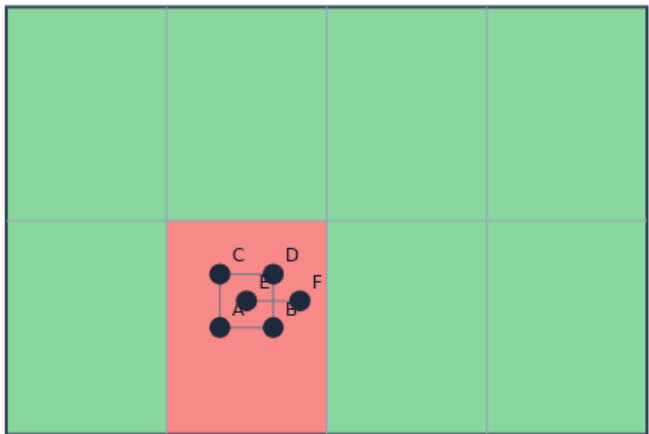
Demand / Cap

Hottest tile

CONGESTION MAP

Hottest tile

Step 2 / 5 · hot · module02-05-congestion-map



Explanation

Argmax over the matrix with fixed scan order for stable goldens.

- Name (i,j)

Center on seed

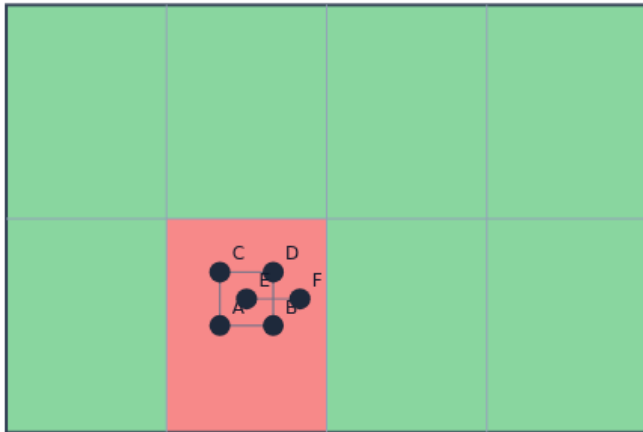
Hottest tile

Read the colors

CONGESTION MAP

Read the colors

Step 3 / 5 · legend · module02-05-congestion-map



Explanation

Hotter colors mean higher congestion; use metrics for exact floats.

- Don't eyeball only

Regression uses numbers

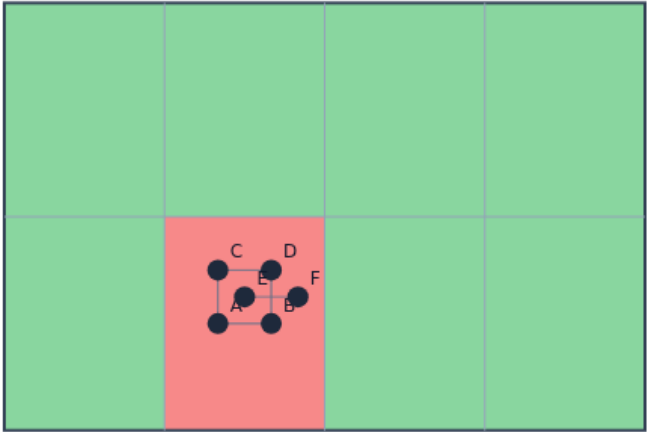
Read the colors

Move the hotspot

CONGESTION MAP

Move the hotspot

Step 4 / 5 · move · module02-05-congestion-map



Explanation

Dragging cells can move which tile is hottest.

- Learner state

Challenges score positions

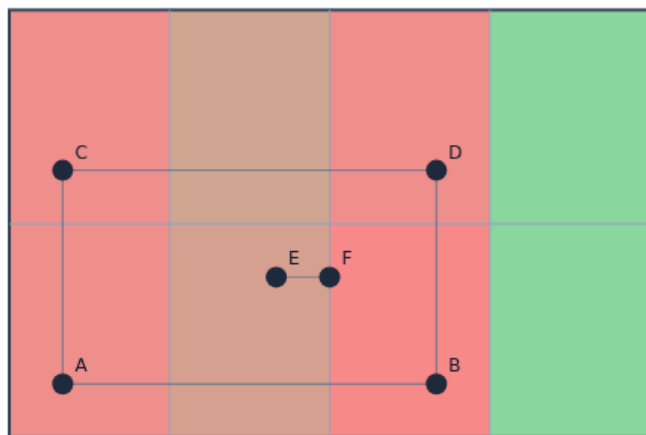
Move the hotspot

Cooler map

CONGESTION MAP

Cooler map

Step 5 / 5 · spread · module02-05-congestion-map



Explanation

Spread placement lowers peak ratios.

- Toward routing

Overflow lab next

Cooler map

Browser lab track

EDA Algorithms Platform

HomeTools

Home / Tools / Congestion heat map

INTERACTIVE TOOL

Congestion heat map

Paint demand/capacity per GCell and find the hottest tile.

Your job: congestion = demand/capacity. Move cells until the hottest GCell is not a center tile — or cool the map below thresholds.

Reset workspace

Challenges0 / 8 cleared

Pick oneIntro] Hottest defined

Pick a challenge and click **Start**. Move cells, then **Check**. Reveal golden is study-only.

Start

Show hint

Check

Next challenge

Stop checking

Idle

A B C D E F ← → ↑ ↓

Reset to starter

Reveal golden (study)

Run feedback

Graph

Reference / metrics

selected: A @ GCell (1,0)
HPWL: 7.0
overflow total: 5.00 · max: 5.00 · count: 1
capacity: 2
view: your placement
hottest: (1.0) @ 3.50

Browser lab starter

learn_congestion — 05 congestion map

Implement track

- Build ``congestion_map(demand, capacity)`` returning the ratio matrix and hottest index
- Print hottest for both placement seeds

Pitfalls

- Dividing by zero capacity
- Pick a fixed scan order

Your turn

- Finish the lab
- Next: overflow metrics compress the map into totals you can regress